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NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of NITTE (Deemed to be University)

First Semester M.Tech (CS&Engg) (CBCS) Degree Examinations

December 2024- January 2025

23CSE101 – ADVANCED DATA STRUCTURES AND ALGORITHMS

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit

Unit – I		Marks	BT*	CO*	PO*
1. a)	With suitable sketch explain the concept of circular queue using dynamic array with the algorithm for handling the queue full condition.	8	L*3	1	2,3
b)	Design a data structure to have a singly linked list in C. Develop a stack data structure using a singly linked list giving corresponding C functions.	6	L3	1	2,3
c)	Explain the concept behind double hashing and show the double hashing for the following example keys : 35,129,36,47,25,26,65,29,16,14 Hash function $h(\text{key}) = \text{key} \% 15$	6	L3	1	2,3
2. a)	Design proper data structure to represent the following 5X4 matrix using circular linked list. Show the links clearly in the sketch. 2 0 0 0 4 0 0 3 0 0 0 0 8 0 0 1 0 0 6 0	8	L3	1	2,3
b)	Design the C functions for a doubly linked list (DLL) to perform the following (i) To create a DLL of n nodes using front insertion. (ii) to insert a node to left of a given node and Delete the given node	6	L3	1	2,3
c)	What do you mean by collision in hashing? How collision is resolved in linear probing method? Show collision resolution by Linear probing for the data given below keys : 35,129,36,48,25,26,65,29,16,14,95 Hash function $h(\text{key}) = \text{key} \% 13$	6	L3	1	2,3
Unit – II					
3. a)	What is a level order traversal in a binary tree? Explain with suitable algorithm and an example. Outline the Inorder and preorder traversal algorithm for a binary tree Construct a binary tree for the given traversals Preorder: + * * / A B C D E Inorder : A / B * C * D + E	8	L3	2	2,3
b)	What is a 2-3 tree? Construct such a 2-3 tree for input 12,56,89,16,25,100,75,55,95,50 showing step by step insertion, giving the technical issues handled to keep tree balanced.	6	L3	2	2,3
c)	Show four types of rotations used in AVL tree to keep tree balanced. Construct an AVL tree for the list 5, 6, 8, 3, 2, 4, 7 by successive insertions, also specify the type of rotations used.	6	L3	2	2,3
4. a)	Define a Binary search tree. Construct a binary search tree showing step by step insertion for the given list 14,15,4,9,7,18,3,5,16,20,9,17 Also show the method of deleting the root key at a time until tree is empty. Give the order of keys deleted.	8	L3	2	2,3

- b) Define B-tree and give it's properties. Show the step by step method of constructing a B-tree of order of $m=4$ for the keys 40, 10, 70, 20, 50, 80, 90, 30, 60, 25, 15, 5, 65.
- c) What is a splay tree? Give its key feature. List out all types of rotations used to Construct a splay tree. Construct a splay tree for the keys 10, 20, 15, 5, 8, 2 commenting on type of rotation used. Also show the splay tree after search a key 10 for that constructed splay tree.

6	L3	2	2,3
6	L3	2	2,3

Unit – III

5. a) Define the Asymptotic notations Big oh, Big Omega and Big theta and explain them by giving suitable examples to each. Compare the following with suitable examples
- (i) little oh with big oh
- (ii) little omega with big omega.
- b) Give ratio theorem for comparing order of growth and how it can be used to represent the algorithm efficiency asymptotically. Give the basic efficiency asymptotic classes and comments on that.
6. a) Explain in brief How you would solve a recurrence relation by recursion method. Construct the recursion tree for the recurrence relation $T(n)=3T(n/4)+cn^2$ where c is a constant > 0 and hence analyze its efficiency
- b) Give the general plan for analyzing the efficiency of a non-recursive algorithms. Show the mathematical analysis of the non-recursive algorithms to state the efficiency of algorithms that
- (i) Determines whether elements in an array are distinct
- (ii) Computes the number of binary digits in the binary representation of a positive decimal integer

10	L3	3	2,3
10	L3	3	2,3
10	L3	3	2,3
10	L3	3	2,3

Unit – IV

7. a) Outline Dijkstra's algorithm. Show the application of Dijkstra's algorithm to the graphs shown in Fig. 7.a1 and 7.a2 and give your conclusions on that.

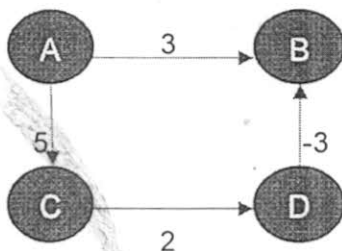


Fig. 7.a1

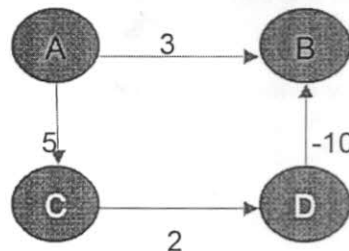
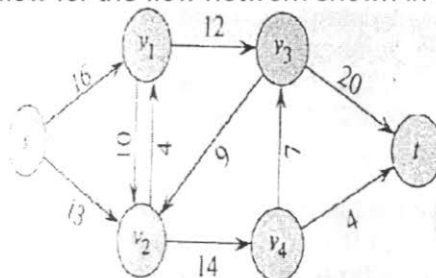


Fig. 7.a2

- b) Explain with algorithm Johnson's algorithm to find the all pair shortest path. Discuss its efficiency and compare with Floyd's algorithm.

10	L3	4	2,3
10	L2	4	2,3

8. a) Define the term maximum flow in a flow network and give a suitable algorithm to compute the maximum flow for the flow network. Find the maximum flow for the flow network shown in Fig.8.a



10	L3	4	2,3
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- b) Define Maximum Bipartite Matching problem. Solve maximum bipartite matching for the following graph (shown in Fig 8.b) having two sets of vertices.

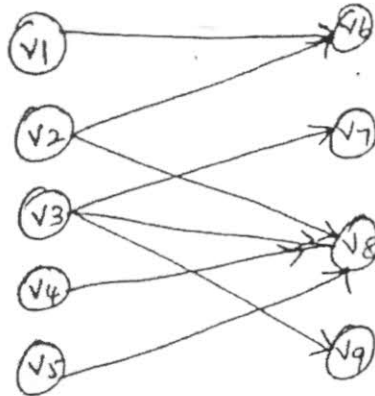


Fig 8. b

10 L3 4 2,3

Unit – V

9. a) Define string matching problem. Give naïve string-matching algorithm and explain how it works. Trace it for a given text acaabc and pattern aab. 10 L3 5 2,3
- b) Give the Rabin-Karp string matching algorithm. Apply this algorithm to find a pattern 31415 in a text 2359023141526739921. 10 L3 5 2,3
10. a) How do you randomize Quick Sort algorithm? Give randomized quick sort algorithm. Compare quick sort algorithm performance with randomized Quick Sort algorithm. 10 L3 5 2,3
- b) Explain in brief how string matching with finite automata works. Apply it for a string matching problem where
Text: abababacaba
Pattern: ababaca 10 L3 5 2,3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
